

Technical Requirement Document

Here's the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001:

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a PySpark job using Delta Live Tables (DLT) to load customer and order data from CSV files to Delta tables, applying schema and data transformations.

Target Cluster Configuration:

- Cluster Name: Data Engineering Cluster
- Databricks Runtime Version: 12.2 LTS
- Node Type: Standard_DS4_v2
- Driver Node: Standard_DS4_v2
- Worker Nodes: 2-4 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Customer Data: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata, CSV, Customer information
- Order Data: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata, CSV, Order information

Target Data Details:

- Customer Delta Table: customer, Delta, Cleansed customer data
- Order Delta Table: order, Delta, Cleansed order data with TotalAmount column

Job Flow / Pipeline Stages:

- Read customer and order data from CSV files
- Apply schema to customer and order data
- Add TotalAmount column to order data
- Load data into customer and order Delta tables

Data Transformations / Business Logic:

- Apply schema: CustId, Name, EmailId, Region for customer and OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order
- Calculate TotalAmount: PricePerUnit Qty

Error Handling and Logging:

- Log errors during data loading and transformation
- Implement retry mechanism for transient errors

TR-DLT-002:**Related Functional Requirement(s): FR-DLT-002**

Objective: Implement data cleansing logic to remove null and duplicate records from customer and order Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Customer Delta Table: customer, Delta, Customer data with potential nulls and duplicates
- Order Delta Table: order, Delta, Order data with potential nulls and duplicates

Target Data Details:

- Cleansed Customer Delta Table: customer, Delta, Customer data without nulls and duplicates
- Cleansed Order Delta Table: order, Delta, Order data without nulls and duplicates

Job Flow / Pipeline Stages:

- Remove null records from customer and order Delta tables
- Remove duplicate records from customer and order Delta tables

Data Transformations / Business Logic:

- Remove null records: filter out rows with null values
- Remove duplicates: drop duplicate rows based on primary keys

Error Handling and Logging:

- Log number of records removed
- Log errors during data cleansing

TR-DLT-003:**Related Functional Requirement(s): FR-DLT-003**

Objective: Create ordersummary table and load data by joining customer and order Delta tables, applying SCD type 2 logic.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Customer Delta Table: customer, Delta, Cleansed customer data
- Order Delta Table: order, Delta, Cleansed order data

Target Data Details:

- ordersummary Table: ordersummary, Delta, Joined customer and order data with SCD type 2 logic

Job Flow / Pipeline Stages:

- Create ordersummary table if it doesn't exist
- Join customer and order Delta tables on CustId
- Apply SCD type 2 logic to ordersummary table

Data Transformations / Business Logic:

- Join customer and order data: inner join on CustId
- SCD type 2 logic: maintain history of changes with StartDate and EndDate

Error Handling and Logging:

- Log errors during data loading and transformation
- Log number of records inserted and updated

TR-DLT-004:**Related Functional Requirement(s): FR-DLT-004**

Objective: Update ordersummary table when there are changes to the customer table, applying SCD type 2 logic.

Target Cluster Configuration: Same as TR-DLT-001**Source Data Details:**

- Customer Delta Table: customer, Delta, Customer data with potential changes
- ordersummary Table: ordersummary, Delta, Existing ordersummary data

Target Data Details:

- Updated ordersummary Table: ordersummary, Delta, Updated ordersummary data with SCD type 2 logic

Job Flow / Pipeline Stages:

- Detect changes to customer table
- Update ordersummary table by making old records inactive and new records active
- Update StartDate and EndDate accordingly

Data Transformations / Business Logic:

- Detect changes: compare current and previous customer data
- SCD type 2 logic: update ordersummary table with new and changed records

Error Handling and Logging:

- Log errors during data update
- Log number of records updated

TR-DLT-005:

Related Functional Requirement(s): FR-DLT-005

Objective: Create customeraggregatespend table and load aggregated data from ordersummary table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- ordersummary Table: ordersummary, Delta, ordersummary data

Target Data Details:

- customeraggregatespend Table: customeraggregatespend, Delta, Aggregated customer spend data

Job Flow / Pipeline Stages:

- Create customeraggregatespend table if it doesn't exist
- Aggregate TotalAmount from ordersummary table by Name and Date
- Load aggregated data into customeraggregatespend table

Data Transformations / Business Logic:

- Aggregate data: group by Name and Date, sum TotalAmount

Error Handling and Logging:

- Log errors during data aggregation and loading
- Log number of records inserted

TR-DLT-006:

Related Functional Requirement(s): FR-DLT-006

Objective: Implement data processing pipeline using Delta Live Tables (DLT).

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement steps 1 to 5 using DLT, ensuring data is processed in the correct order and with required transformations

Data Transformations / Business Logic:

- Orchestrate data processing pipeline using DLT

Error Handling and Logging:

- Log errors during pipeline execution
- Log pipeline performance metrics